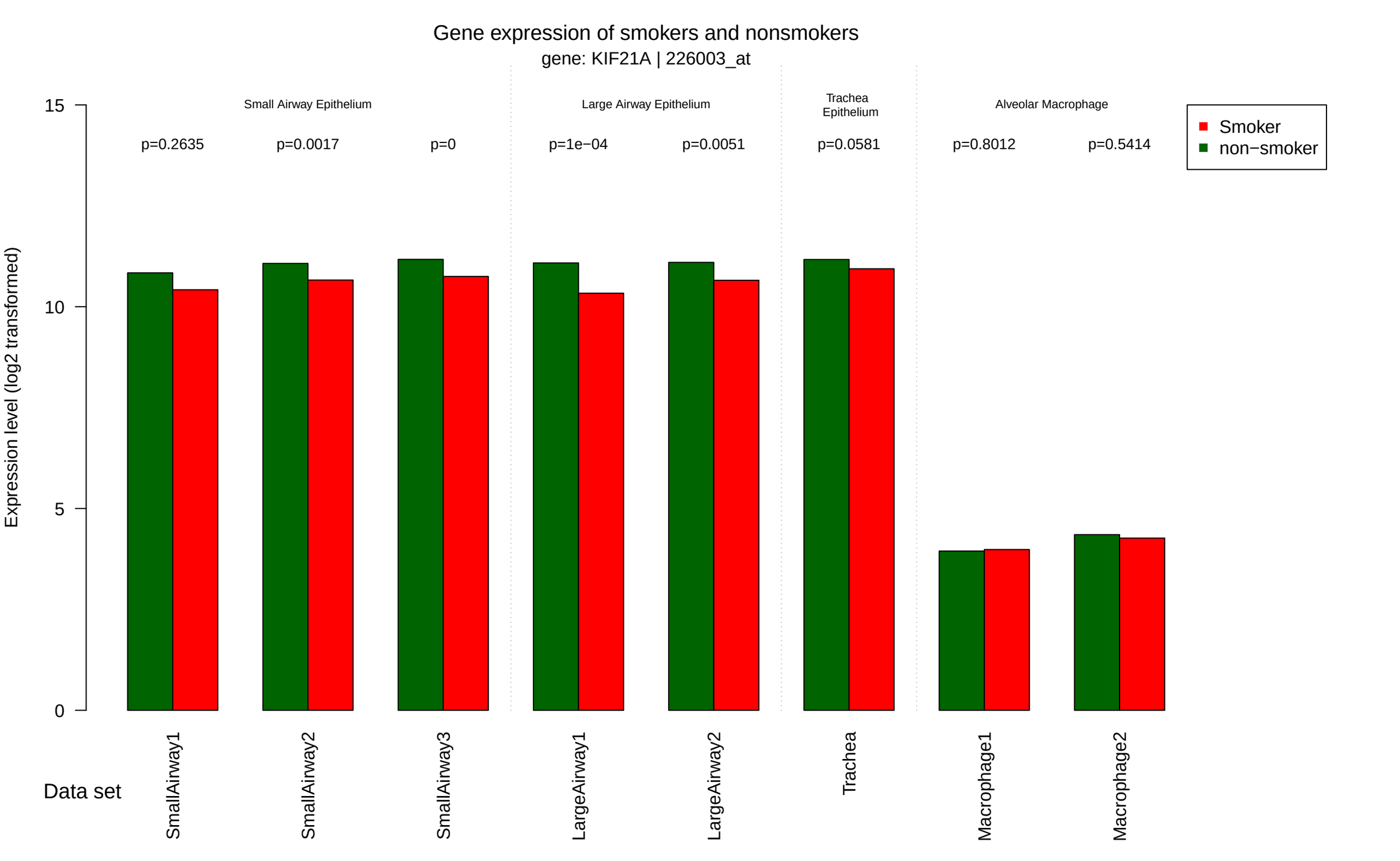
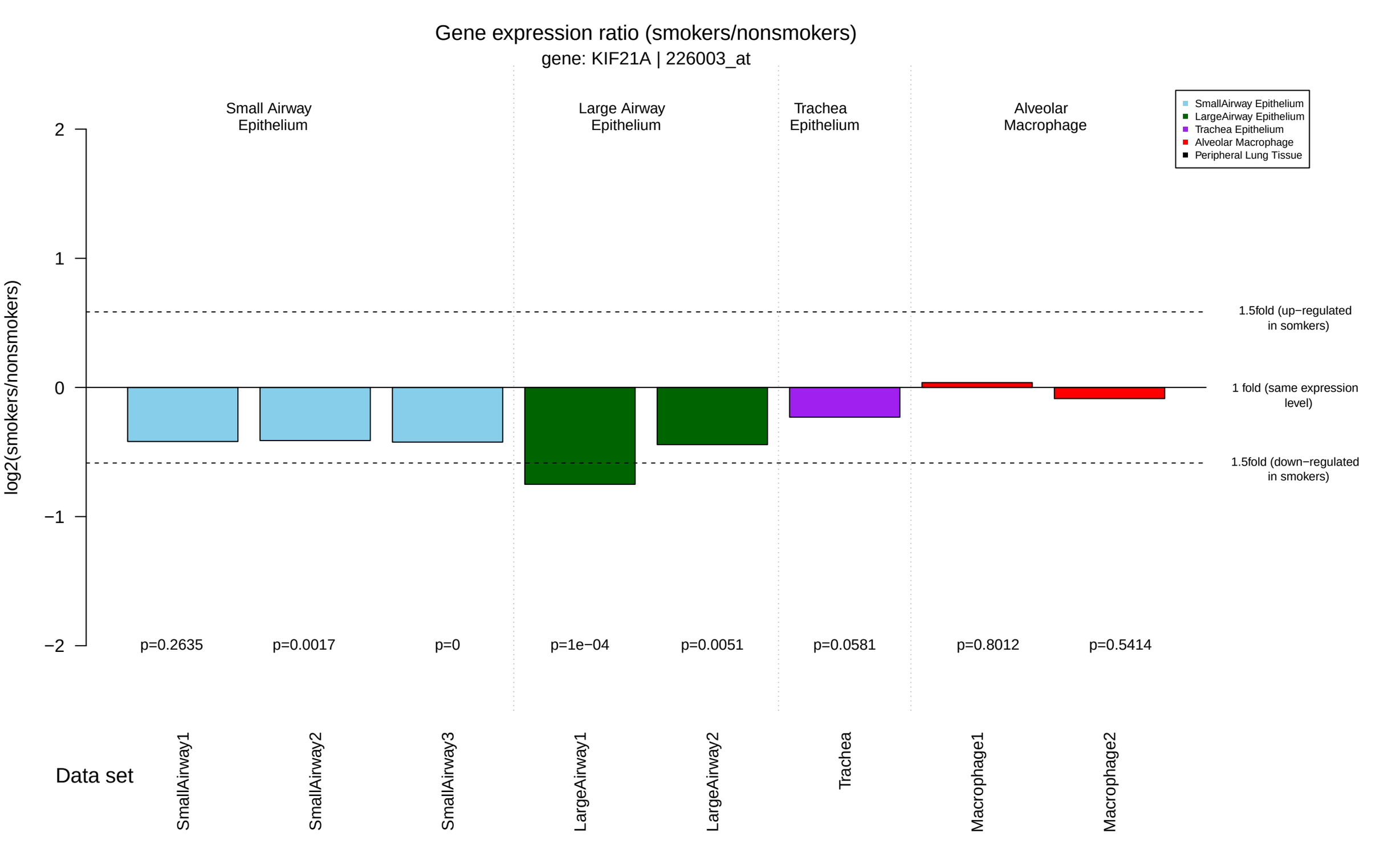
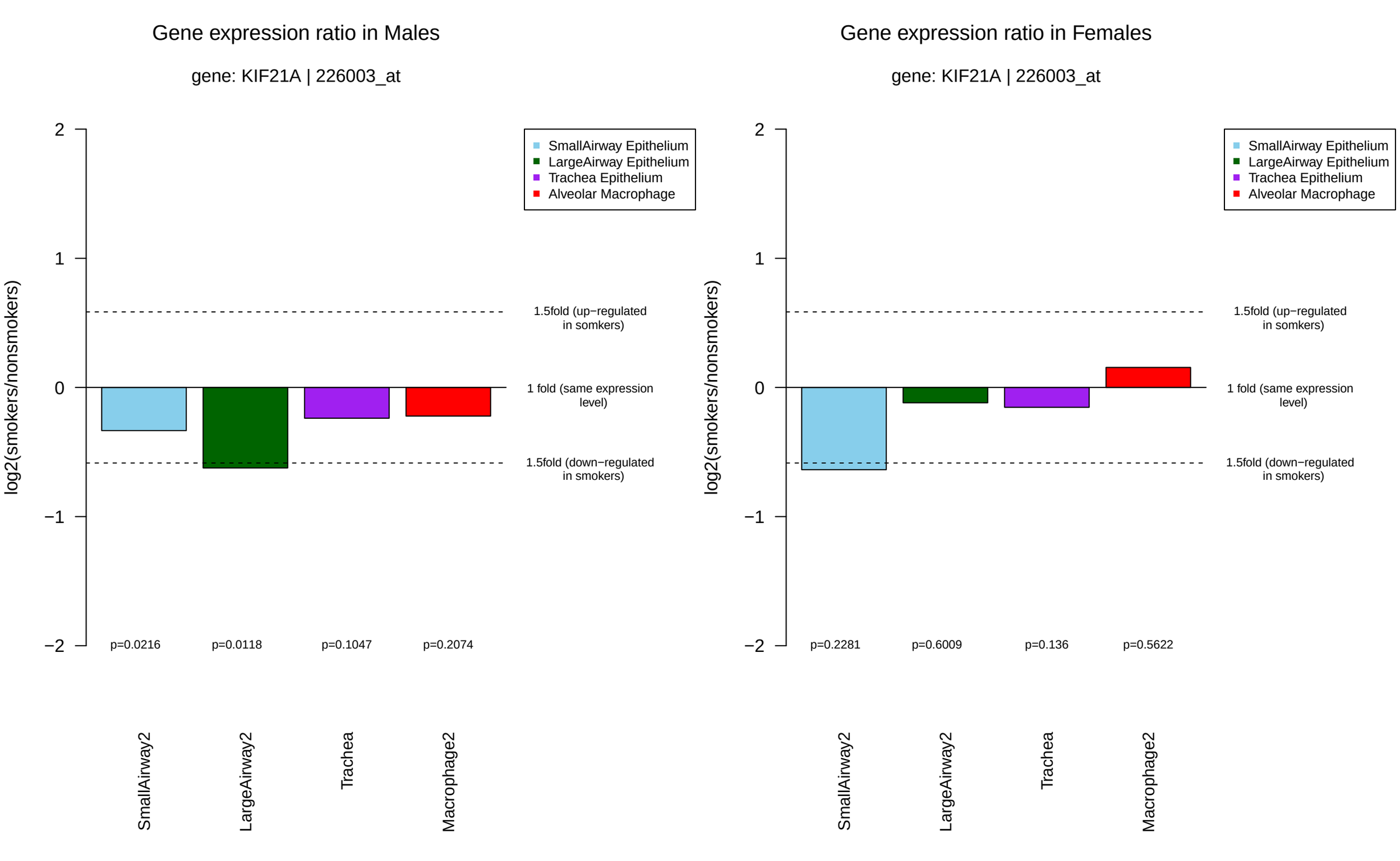
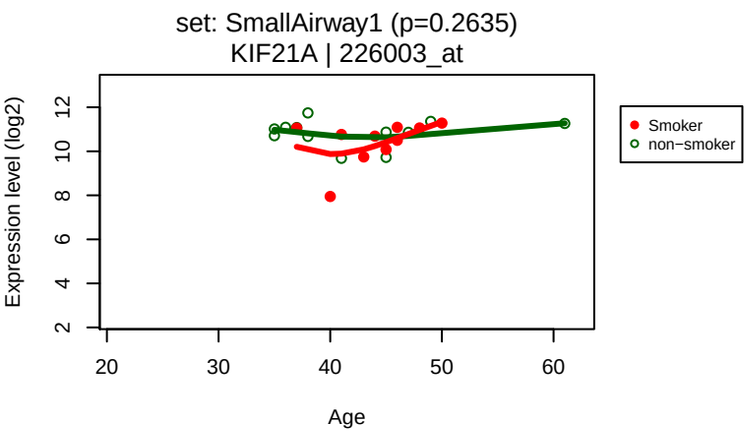
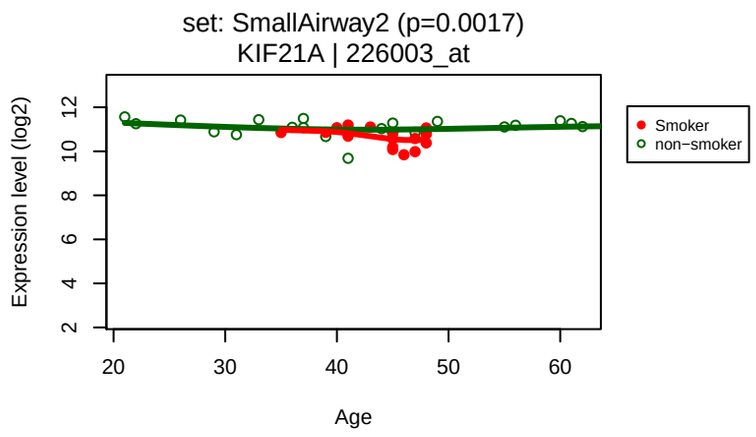
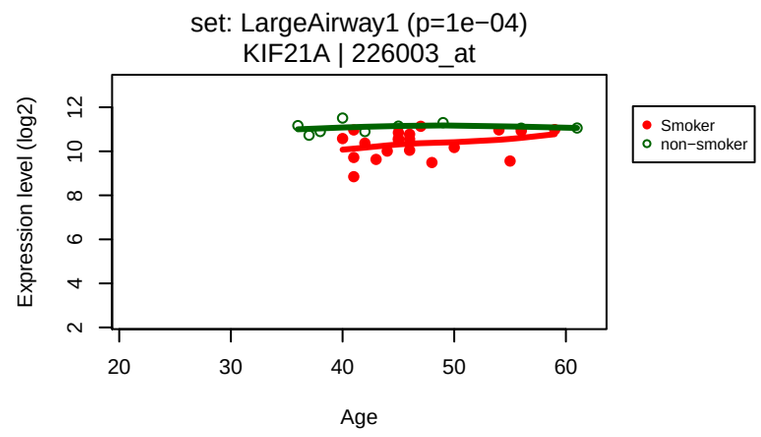

Gene expression of smokers and nonsmokers
gene: KIF21A | 226003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2635
p=0.0017
p=0
p=1e−04
p=0.0051
p=0.0581
p=0.8012
p=0.5414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

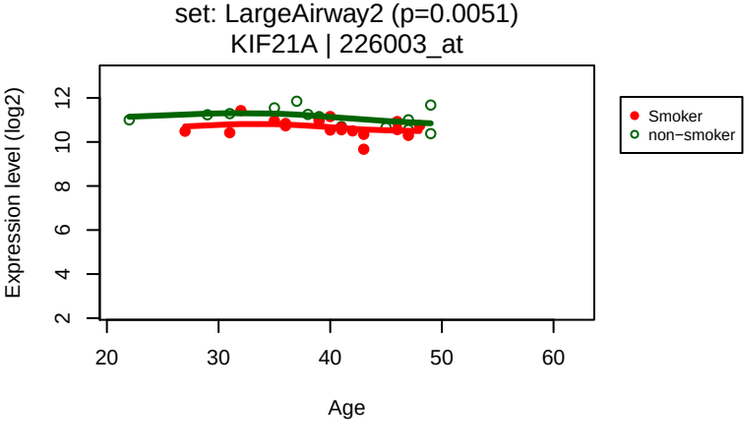

Gene expression ratio (smokers/nonsmokers)
gene: KIF21A | 226003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2635
p=0.0017
p=0
p=1e−04
p=0.0051
p=0.0581
p=0.8012
p=0.5414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

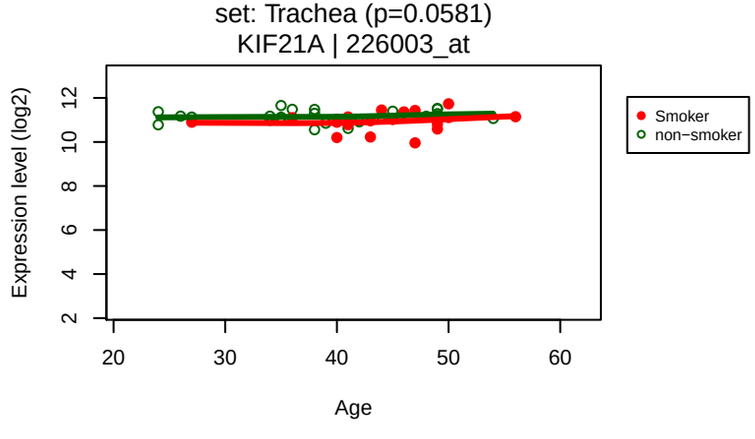

Gene expression ratio in Males
gene: KIF21A | 226003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0216
p=0.0118
p=0.1047
p=0.2074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIF21A | 226003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2281
p=0.6009
p=0.136
p=0.5622
SmallAirway2
LargeAirway2
Trachea
Macrophage2

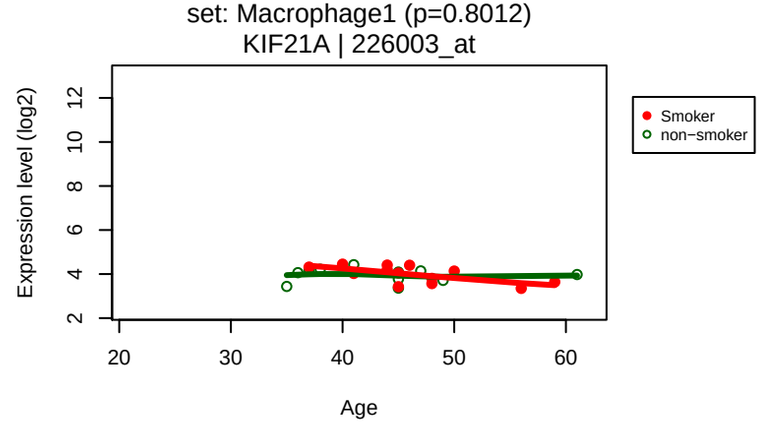

set: SmallAirway1 (p=0.2635)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker

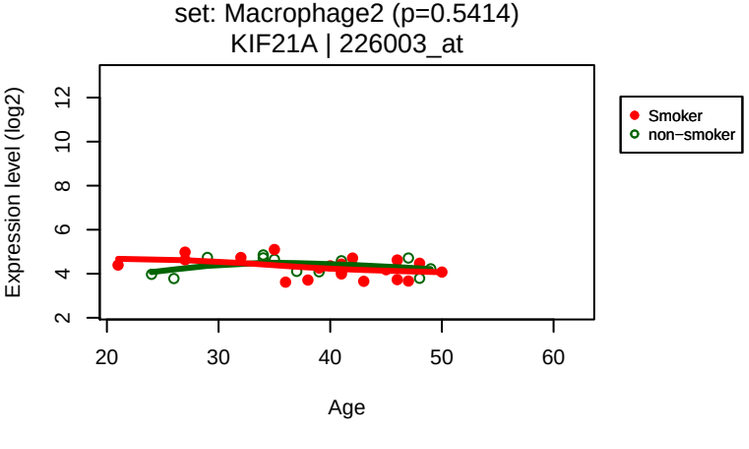

set: SmallAirway2 (p=0.0017)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker

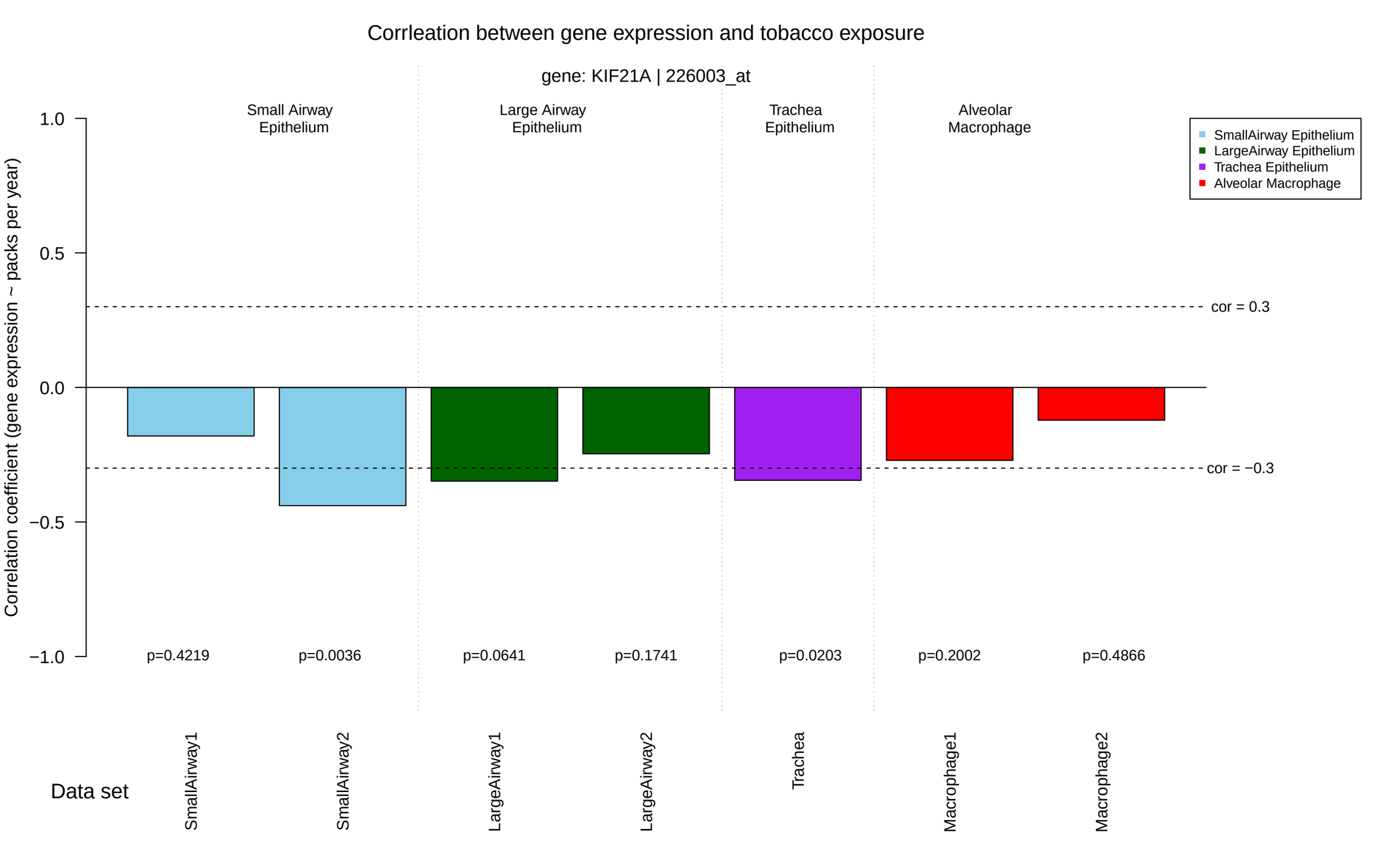

set: LargeAirway1 (p=1e−04)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0051)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0581)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8012)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5414)
KIF21A | 226003_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIF21A | 226003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4219
p=0.0036
p=0.0641
p=0.1741
p=0.0203
p=0.2002
p=0.4866
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2